

5 Gb/s NRZ SerDes Receiver in 65-nm CMOS

Sourojit Mazumder, Guy Robbins, Jack Chang, Siddharth Vadyalam, Ashton Billings, Pete Sanguanchua

Abstract—High-speed serial links have become fundamental to modern computing and communication systems, enabling efficient and fast data transfer across printed circuit boards, cables, and backplanes while minimizing pin count and power consumption. The widely adopted serializer/deserializer (SerDes) architecture addresses many of the fundamental limitations of wide parallel interfaces, including excessive I/O pin requirements, clock skew between parallel lanes, increased EMI, and higher power consumption. In this project, we propose the design of a 5 Gb/s NRZ SerDes receiver. At this data rate, channel loss, ISI, and timing uncertainty can significantly degrade signal integrity. Therefore, emphasis will be placed on demonstrating the impact of channel loss and the effectiveness of equalization techniques through eye diagram analysis and BER estimation. As an advanced feature, we aim to implement adaptive equalization techniques that enable self-tuning of our chip's equalization blocks; this is extremely useful at high data rates as the channel properties will vary depending on the interface used.

I. PROJECT MOTIVATION

Using parallel data buses becomes impractical at higher speeds due to several fundamental limitations. Wide parallel interfaces require a large number of I/O pins, increasing chip area and package complexity, and suffer from higher power consumption and EMI. Additionally, maintaining precise timing alignment across multiple parallel lanes becomes increasingly difficult due to clock skew, process variations, etc. These challenges motivated the transition to SerDes architectures, where parallel data is converted into a high-speed serial stream at the transmitter (TX) and reconstructed at the receiver (RX).

However, serial transmission introduces new challenges associated with high-speed signaling over bandwidth-limited channels. PCB traces and cables exhibit frequency-dependent loss, reflections, and dispersion, which distort the transmitted waveform and introduce intersymbol interference (ISI). Additionally, at our target data rate of 5 Gb/s, the bit period is only 200 ps, leaving a narrow timing margin for accurate signal detection. Furthermore, SerDes links do not transmit a dedicated clock signal, which requires the RX to recover timing information from the incoming data stream.

To address these challenges, our proposed receiver will incorporate key blocks used in modern SerDes designs, including a full-rate Clock and Data Recovery (CDR) loop integrated with an analog front-end comprising a Continuous-Time Linear Equalizer (CTLE) and a Decision Feedback Equalizer (DFE). A deserializer block will convert the serialized data to the desired parallel output.

A 5 Gb/s NRZ SerDes receiver strikes a good balance between complexity and feasibility for a semester-long effort. Given the complexity inherent in the receiver itself, we will only be designing the RX. The TX will be emulated using a Bit Error Rate Tester (BERT), discussed later in Section IV.

II. TARGET SPECIFICATIONS

The list below is a good starting point for our project. As our work progresses, it is possible additional specs may be added to the list, while some of the given specs may need to be modified. The BER spec is very important to our design, as the RX must reliably maintain good signal integrity.

- Data Rate: > 5 Gb/s
- Signaling scheme: Non-return-to-zero (NRZ), using differential signaling
- Bit Error Rate (BER): $< 10^{-12}$
- Power Consumption: < 20 mW
- Supply Voltage: 1 V
- Area: $< 1\text{mm}^2$
- Serialization Factor: 1 serial lane to 4 parallel lines

III. SYSTEM ARCHITECTURE

The top-level block diagram of our proposed chip is illustrated in Figure 1. The individual blocks are discussed in the remainder of this section along with any necessary sub-block diagrams.

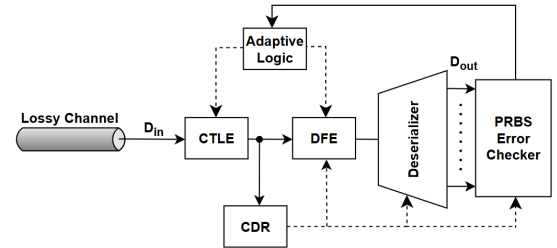


Fig. 1: SerDes top-level block diagram.

A. Continuous-Time Linear Equalizer (CTLE)

As Figure 2 suggests, all real channels have some frequency-dependent loss profile. At the low GHz-range frequencies we want to operate our chip at, a loss of, at minimum, a few dB is essentially guaranteed. The CTLE, as the name suggests, is a linear circuit designed to perform the first stage of equalization of the signal. It accomplishes this by providing sufficient gain at the high frequency components of the signal that are typically attenuated by the channel, ideally equalizing the signal over all frequencies within bandwidth.

If we have a fixed channel, then we can measure the loss profile (S-parameters) as a function of frequency and design our CTLE with the exact profile in mind. However, if the characteristics of the channel are unknown to us (which is often the case practically), then it would be necessary to make the characteristics of the CTLE programmable via the chip's pin(s). This includes the CTLE's bandwidth to ensure

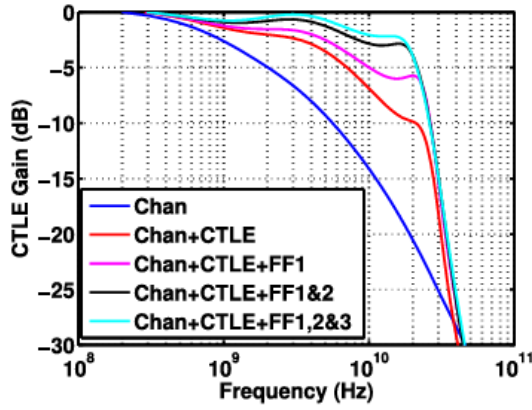


Fig. 2: Some possible frequency response plots [3] for the channel-CTLE cascade. The CTLE helps reduce the roll-off due to the lossy channel.

the equalization is at least above the Nyquist rate and its peaking gain to ensure good signal integrity for the subsequent components. The CTLE architecture is illustrated in Figure 3.

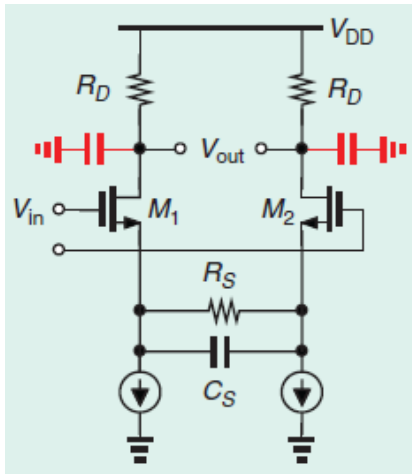


Fig. 3: Transistor-level schematic for the CTLE. The degeneration capacitors can be implemented as a cap bank (caps and switches) to allow for programming the CTLE.

1) *Advanced Feature*: Provided the fundamental features of the SerDes have been completed first, a potential advanced feature that may be implemented would be a form of adaptive equalization. As illustrated in Figure 1, one way to accomplish this is through the use of an on-chip error checker. As we discuss in more detail in Section IV, the input to the channel is generated using a Bit Error Rate Tester (BERT). This equipment can generate a pseudorandom binary sequence (PRBS), a quasi-random sequence of bits, and send it to our chip via a PCB trace. Since PRBS is deterministic in nature, the on-chip checker can compare the input bits and output bits and send an error (feedback) signal to our adaptive logic circuitry. Thus, if the initial BER is too high, the CTLE and DFE blocks will be automatically tuned until an optimal BER is achieved.

B. Decision Feedback Equalizer (DFE)

The DFE is a non-linear equalizer designed to mitigate post-cursor inter-symbol interference (ISI). Unlike linear equalization, which often suffers from high-frequency noise amplification and channel notch effect, the DFE improves signal integrity through subtracting the weighted influence of the previously detected bits. For a 5 Gb/s application, the primary design constraint is closing this critical feedback path within a single 200 ps unit interval (UI). To achieve this while simultaneously easing high-speed clock distribution, we chose a full-rate direct-feedback architecture that operates on the clock recovered in the CDR.

1) *Summation Node*: The analog summer acts as the primary junction for subtracting the post-cursor ISI of the previous bit from the current bit. It takes the continuous incoming signal from the CTLE and combines it with the equalization feedback from the h_1 tap. By performing this operation in the current domain, the block is able to achieve high-speed analog subtraction prior to digitization. The transistor-level implementation selected is shown in Figure 4. The number of differential pairs would determine the number of tap weights the DFE could use to subtract the post-cursor ISI, which would be hand-tuned during post-silicon validation to achieve low BER.

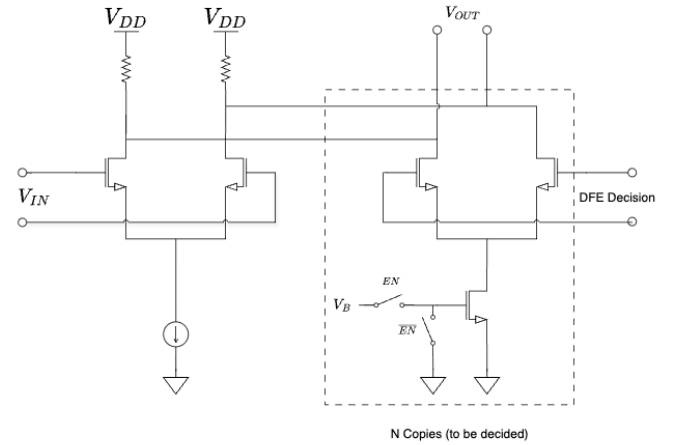


Fig. 4: Current Summer Architecture

2) *Slicer*: The slicers act as high-speed clocked comparators and latches that digitize the analog output of the summation node into a clean digital signal and hold that decision at the output node. The slicer output feeds directly into both the deserializer and feedback tap (h_1). The StrongARM latch architecture is chosen for its simplicity and high-speed operation.

3) *Feedback Tap*: The feedback tap is responsible for generating the h_1 equalization weight, corresponding to the ISI tail that needs to be subtracted from the incoming signal. It functions as a 1-bit DAC that takes the previous bit (output of SR latch) and converts it into a weighted analog current.

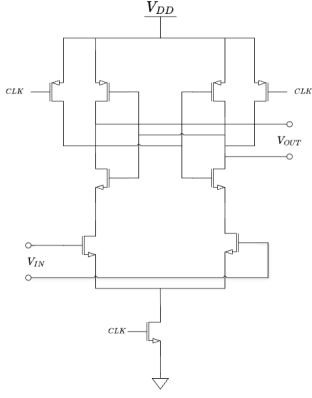


Fig. 5: StrongARM Latch Architecture

This current is fed directly to the summation node, where it is subtracted from the current bit, forming a feedback loop.

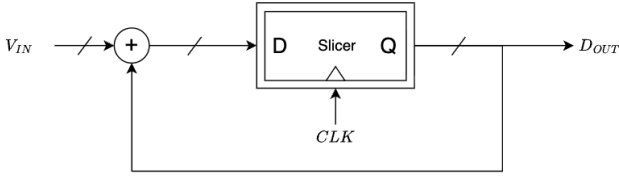


Fig. 6: DFE Direct Feedback Block Diagram

C. Clock and Data Recovery (CDR)

The CDR circuit is designed to extract the timing information from the incoming data stream and generate a synchronized clock for data sampling. The desired output frequency of this circuit is 5 GHz, the same as the input data rate. The design constraints of this module include the output stability, as the DFE and deserializer circuits' performance relies on the output of this module. Our architecture consists of a Phase Detector (PD), Charge Pump (CP), Loop Filter (LP), and Voltage Controlled Oscillator (VCO).

1) *Hogge Phase Detector (HPD)*: The phase detector is the first component of the CDR, taking the inputted bitstream and VCO clock as inputs. Using these two data streams, it calculates the phase difference to feed into the next stage, the CP. In our architecture we choose to define a linear PD, which produces a DC output signal proportional to the phase difference. This circuit is composed of a positive and a negative edged flip flop, and two XOR gates. The first flip flop produces a pulse that is proportional to the input data, while the second produces a reference half period pulse to move the proportional pulse to be centered at the origin.

2) *Charge Pump (CP) and Loop Filter (LF)*: The Charge Pump takes the proportional pulse, and the half period pulse to generate sourcing and sinking currents. The difference in duration between these pulses produces a net charge in the

loop filter proportional to the detected phase error. Near lock, this net charge will be near 0. The loop filter takes the charge pump output and integrates it with a second order loop filter to act as a control voltage for the VCO. The filter includes a series resistor and capacitor for stability in a second order system and another capacitor in parallel for ripple suppression caused by the charge pump.

3) *Ring VCO*: The VCO is implemented as a multi-stage ring of inverters. The block will be able to support discrete and continuous tuning of the frequency to get the frequency within the desire frequency range. Coarse tuning will be implemented through discretely increasing load capacitance through a thermometer scheme, allowing any PVT corner to be tuned towards our desired frequency. Once within locking range of the CDR, fine tuning will be implemented through variable resistance in each inverter.

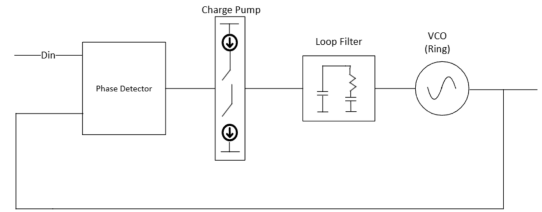


Fig. 7: Architectural Block Diagram of the CDR.

D. Deserializer

The deserializer is the final stage of the RX architecture. This converts the 5 Gb/s serial stream into low-frequency parallel buses for digital processing. Due to the system's implementation as a full-rate architecture, the deserializer is reliant upon a cascaded 1:2 demultiplexing tree to attain the overall 1:4 demux ratio. This reduces each output bit's data rate to 1.25 Gb/s per lane and thus keeping the aggregate data rate to 5 Gb/s.

1) *Tree Architecture*: The full-rate CDR provides a retimed 5-Gb/s data stream and a synchronized 5-GHz clock. Here we can implement a divide by two circuit to generate a 2.5 GHz clock from the CDR recovered clock. The first stage of the deserializer uses this new clock and samples at opposite phases to generate two separate even and odd data bits into two 2.5 Gb/s streams, performing a 1-2 demux operation. This operation can be repeated, with another divider to form a 1.25 GHz clk, with each of the two 2.5 GHz signals being further demultiplexed to form 4 1.25 GHz parallel signals.

2) *Clock synchronization*: The divide-by-2 toggle flip-flop will be implemented by standard logic gates. The clocks running the deserializer must maintain a 50% duty cycle to ensure stability for the 1.25 GHz reliant stage of the demultiplexing tree, further addressing the cruciality of the CDR performance. Skew minimization and phase alignment ensure the system runs reliably.

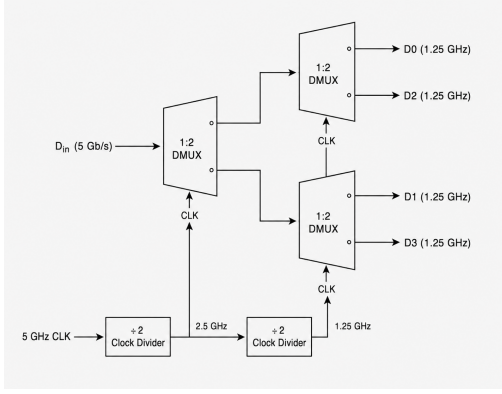


Fig. 8: Architectural Block Diagram of the Deserializer

IV. TESTING PLAN

The primary method of testing involves using a Bit Error Rate Tester (BERT) to measure the BER of our SerDes receiver. The proposed block diagram for the testing setup is illustrated in Figure 9. The first component of the test setup would be the BERT itself, which can be programmed to generate a pattern of bits (pseudorandom binary sequence (PRBS)) and send it to our chip via a channel (PCB). The BERT can then compare the output bits of our chip with the original bit sequence (error analysis).

Once the data has been deserialized by our chip, a spectrum analyzer and scope can be used to analyze performance characteristics like the clock quality, BER, data rate, power consumption, and eye diagram.

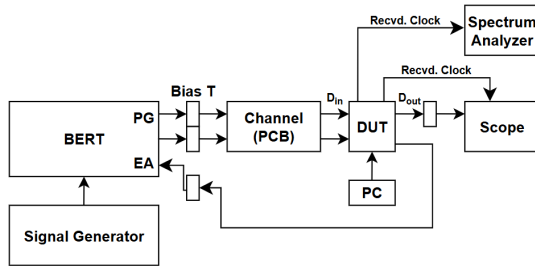


Fig. 9: SerDes test setup

V. ROLES

This may change depending on the needs of each block as our project progresses.

- **Continuous-Time Linear Equalizer (CTLE):** Siddharth Vadyalam, Sourojit Mazumder
- **Decision Feedback Equalizer (DFE):** Guy Robbins, Pete Sanguanchua
- **Clock and Data Recovery (CDR):** Ashton Billings, Jack Chang, Sourojit Mazumder
- **Deserializer (DEMUX):** Siddharth Vadyalam, Sourojit Mazumder

VI. PROJECT PROGRESS

The following section details progress made since the initial proposal. The behavioral model is very close to being complete, with just the CDR left to be completed due to the issue with Verilog-AMS simulation in the 427 server. The remaining tasks and timeline are outlined in Section VII.

A. Behavioral Model

In order to develop a better understanding of the ideal behavior of our SerDes receiver as well as to have a reliable golden model reference during our design, we have developed a behavioral model for our receiver. This model was developed and refined to meet the specifications outlined previously and was executed using a Verilog-A simulation model, allowing it to integrate directly into the Cadence Virtuoso toolchain. This framework supports isolated subsystem verification by allowing the co-simulation of ideal block models with transistor-level blocks. At present, the CTLE and DFE blocks have been successfully modeled and await testing, while the CDR block is under active development due to issues with the Verilog-AMS simulation features in the 427 server.

1) *Channel Model:* To create a relatively accurate channel model, we used a lumped element circuit model for a 1-inch microstrip copper trace on FR4 board that was developed in [10]. Our final channel model was developed by cascading 12 such models to more accurately reflect a plausible 12-inch channel length.

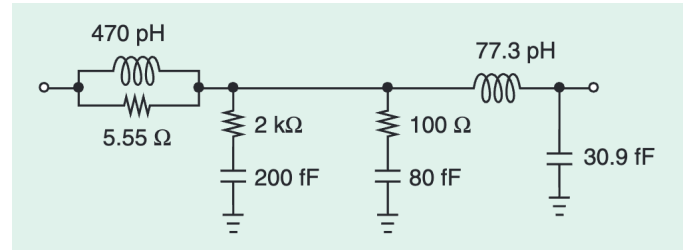


Fig. 10: 1-inch PCB trace model [10]

2) *CTLE Model:* To model an ideal CTLE, we have utilized a transfer function with two poles and one zero. This reflects the expected ideal linear behavior of a differential-pair-based CTLE. The magnitude response of the CTLE is illustrated with a zero at 292 MHz and the poles at 875 MHz and 3 GHz (estimated output pole).

After cascading the channel model with the CTLE model, we obtained the cascaded pulse response to determine the ISI that needs to be compensated by the DFE. Denoting the main cursor to be the peak of the pulse, the first post-cursor of the cascaded channel is roughly around 13%. This simulation provides a rough tap weight target needed for the DFE design.

3) *Summing Node:* The summing node model takes in two inputs: one from the CTLE output and one from the DFE feedback loop, applies proper gain to the two paths, and outputs the resulting summation to the differential node as the input to the slicer stage. This module still requires proper

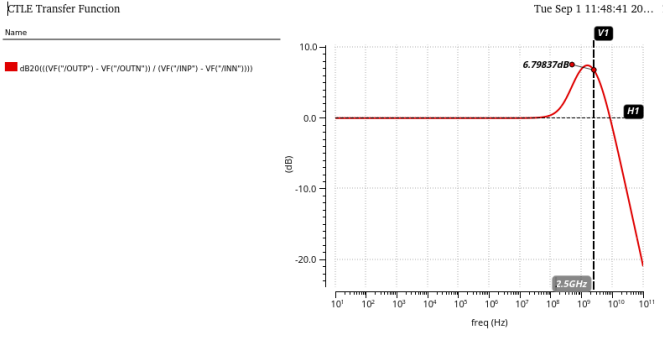


Fig. 11: CTLE Model Frequency Response

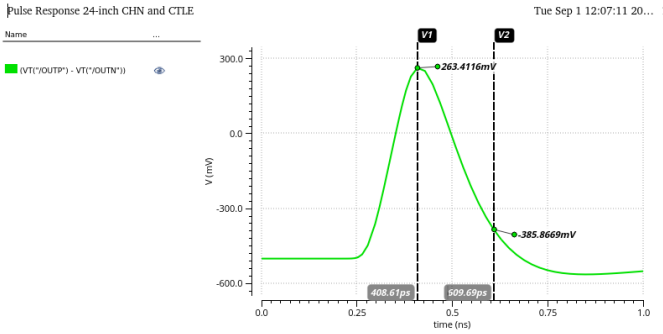


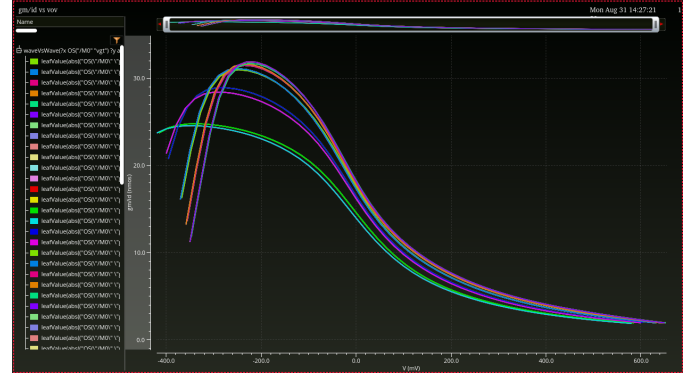
Fig. 12: Pulse Response of the cascaded channel and CTLE

gain tuning to properly open the eye and lower the BER of the resulting output.

4) *Slicer*: The slicer is modeled as a clocked comparator. At the rising edge of the clock, the output of the slicer would determine the output based on the differential input to the slicer.

B. Device Characterization

The given TSMC 65nm process node contains several flavors of the same transistors. After narrowing down the scope of potential transistors based on our spec requirements, we decided to utilize the standard and low threshold voltage transistors as well as their deep n-well variations to minimize noise from substrate and body effects affecting the transistors. Following this, we performed simulations on the individual transistors to obtain various characterization curves which will be used in the initial hand calculations and transistor sizing in the schematic. The main motivation behind this exercise was to overcome the inaccuracy of square law models, which do not account for non-idealities such as short channel effects or velocity saturation. By creating these curves, we essentially created lookup tables that will allow us to avoid trial-and-error sizing during schematic implementation. Most important out of these curves are I_d and g_m/I_d vs V_{ov} , which allow us to use the g_m/I_d methodology. Many other curves were generated against V_{ov} including g_m , I_d/W , and $g_m R_{out}$, in addition to a V_{th} vs L curve, which will all aid in the design process.

Fig. 13: g_m/I_d vs V_{ov} Characterization Curve

VII. UPDATED TIMELINE

Overall, we are close to what we outlined in our initial proposal. The behavioral model is almost complete and schematic design has been started. We aim to finish schematics before the last week of September and use the remaining two months for layout. Each block will be laid out and re-simulated with its extracted parasitics to verify functionality. Afterwards, the full chip will be laid out and again re-simulated after PEX. At each stage in both schematic and layout (block and chip-level design), corners will be fully tested to catch design issues early.

Phase 1: Behavioral Modeling (Aug. - Early Sept.)

Completion of system-level modeling using SystemVerilog. This phase establishes the architectural parameters for loop bandwidth, equalization levels, etc. before moving to transistor-level design.

Phase 2: Schematic Design & Block Testing (Early Sept. - Mid Sept.)

Design and simulation of individual blocks (CTLE, DFE, CDR, and DEMUX). Focus will be on achieving target performance across PVT corners and ensuring target specs are met.

Phase 3: System-Level Integration (Mid Sep. - Late Sep.)

Integration of individual blocks into a complete receiver top-level schematic. This phase focuses on timing closure between the CDR clock and the DFE feedback loop, as well as verifying global biasing and power distribution.

Phase 4: Physical Implementation / Layout (Oct.)

Transistor-level layout. Key focus areas include symmetric routing for the differential pairs in the VCO and CTLE, and careful placement of the loop filter capacitor to minimize parasitic coupling.

Phase 5: Post-Layout Extraction & Verification (Nov.)

Parasitic extraction and back-annotated simulations. This period accounts for the significant simulation time required to verify that the integrated system meets the desired BER and data rate targets after accounting for layout parasitics.

Phase 6: Post-Silicon Validation (Spring 2027)

This phase focuses on the PCB and hardware bring-up of the testing setup described in Section IV.

VIII. GROUP EXPERIENCE

Sourojit Mazumder - B.S. ECE

- Relevant Coursework: ECE 483, 546, 425, 482, 453, 385
- Experience: Analog IC Design Intern at Infineon Technologies, Device Modeling Intern at Infineon
- Technical Strength: Virtuoso, ADS, Analog Design

Guy Robbins - B.S. ECE

- Relevant Coursework: ECE 483, ECE 482, ECE 453, ECE 425
- Experience: Hardware Design Intern at Marvell Semiconductor
- Technical Strength: Cadence Virtuoso, Analog Circuit Design

Jack Chang - B.S. ECE

- Relevant Coursework: ECE 483, ECE 425, ECE 444, ECE 486
- Experience: FPGA Design Intern, Altera
- Technical Strength: Virtuoso, Circuit design, System Modeling

Siddharth Vadyalam - B.S. ECE

- Relevant Coursework: ECE 483, ECE 425, ECE 444, ECE 486
- Experience: Silicon Solutions Engineering Intern at NVIDIA
- Technical Strength: Cadence Virtuoso, Analog Circuit Design

Ashton Billings - M.S. ECE

- Relevant Coursework: ECE 483, ECE 482, ECE 411, ECE 425
- Experience: Analog IC Design Intern at Texas Instruments
- Technical Strength: Analog Circuit Design, Cadence Virtuoso

Pete Sanguanchua - B.S. ECE

- Relevant Coursework: ECE 482, ECE 385, ECE 453
- Experience: Analog IC Design Intern at Analog Devices, High-speed Electrical Validation Intern at Astera Labs
- Technical Strength: Cadence Virtuoso, Circuit Modeling, High-speed IO Testing

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